

Embedding#

<https://pytorch.org/docs/stable/generated/torch.nn.Embedding.html#torch.nn.E>

Description:

A simple lookup table that stores embeddings of a fixed dictionary and size. This module is often used to store word embeddings and retrieve them using indices. The input to the module is a list of indices, and the output is the corresponding word embeddings.

`num_embeddings` (int) – size of the dictionary of embeddings
`embedding_dim` (int) – the size of each embedding vector
`padding_idx` (int, optional) – If specified, the entries at `padding_idx` do not contribute to the gradient; therefore, the embedding vector at `padding_idx` is not updated during training, i.e. it remains as a fixed “pad”. For a newly constructed Embedding, the embedding vector at `padding_idx` will default to all zeros, but can be updated to another value to be used as the padding vector.

Function Signature

```
class torch.nn.Embedding(num_embeddings, embedding_dim,
padding_idx=None, max_norm=None, norm_type=2.0,
scale_grad_by_freq=False, sparse=False, _weight=None,
_freeze=False, device=None, dtype=None)[source]#
```

Parameters

Variables

Shape:

```
classmethod from_pretrained(embeddings, freeze=True,
padding_idx=None, max_norm=None, norm_type=2.0,
scale_grad_by_freq=False, sparse=False)[source]#
```

Code Examples

```
n, d, m = 3, 5, 7
embedding = nn.Embedding(n, d, max_norm=1.0)
W = torch.randn((m, d), requires_grad=True)
idx = torch.tensor([1, 2])
a = (
    embedding.weight.clone() @ W.t()
) # weight must be cloned for this to be differentiable
b = embedding(idx) @ W.t() # modifies weight in-place
out = a.unsqueeze(0) + b.unsqueeze(1)
loss = out.sigmoid().prod()
loss.backward()
```

```

>>> # an Embedding module containing 10 tensors of size 3
>>> embedding = nn.Embedding(10, 3)
>>> # a batch of 2 samples of 4 indices each
>>> input = torch.LongTensor([[1, 2, 4, 5], [4, 3, 2, 9]])
>>> embedding(input)
tensor([[[[-0.0251, -1.6902,  0.7172],
          [-0.6431,  0.0748,  0.6969],
          [ 1.4970,  1.3448, -0.9685],
          [-0.3677, -2.7265, -0.1685]],
         [[ 1.4970,  1.3448, -0.9685],
          [ 0.4362, -0.4004,  0.9400],
          [-0.6431,  0.0748,  0.6969],
          [ 0.9124, -2.3616,  1.1151]]]])

>>> # example with padding_idx
>>> embedding = nn.Embedding(10, 3, padding_idx=0)
>>> input = torch.LongTensor([[0, 2, 0, 5]])
>>> embedding(input)
tensor([[[[ 0.0000,  0.0000,  0.0000],
          [ 0.1535, -2.0309,  0.9315],
          [ 0.0000,  0.0000,  0.0000],
          [-0.1655,  0.9897,  0.0635]]]])

>>> # example of changing `pad` vector
>>> padding_idx = 0
>>> embedding = nn.Embedding(3, 3, padding_idx=padding_idx)
>>> embedding.weight
Parameter containing:
tensor([[ 0.0000,  0.0000,  0.0000],
        [-0.7895, -0.7089, -0.0364],
        [ 0.6778,  0.5803,  0.2678]], requires_grad=True)
>>> with torch.no_grad():
...     embedding.weight[padding_idx] = torch.ones(3)
>>> embedding.weight
Parameter containing:

```

```
tensor([[ 1.0000,  1.0000,  1.0000],
        [-0.7895, -0.7089, -0.0364],
        [ 0.6778,  0.5803,  0.2678]], requires_grad=True)
```

```
>>> # FloatTensor containing pretrained weights
>>> weight = torch.FloatTensor([[1, 2.3, 3], [4, 5.1, 6.3]])
>>> embedding = nn.Embedding.from_pretrained(weight)
>>> # Get embeddings for index 1
>>> input = torch.LongTensor([1])
>>> embedding(input)
tensor([[ 4.0000,  5.1000,  6.3000]])
```

Notes & Warnings

NOTE

Note Keep in mind that only a limited number of optimizers support sparse gradients: currently it's `optim.SGD` (CUDA and CPU), `optim.SparseAdam` (CUDA and CPU) and `optim.Adagrad` (CPU)

NOTE

Note When `max_norm` is not `None`, `Embedding`'s forward method will modify the weight tensor in-place. Since tensors needed for gradient computations cannot be modified in-place, performing a differentiable operation on `Embedding.weight` before calling `Embedding`'s forward method requires cloning `Embedding.weight` when `max_norm` is not `None`. For example: `n, d, m = 3, 5, 7` `embedding = nn.Embedding(n, d, max_norm=1.0)` `W = torch.randn((m, d), requires_grad=True)` `idx = torch.tensor([1, 2])` `a = (embedding.weight.clone() @ W.t())` # weight must be cloned for this to be differentiable `b = embedding(idx) @ W.t()` # modifies weight in-place `out = a.unsqueeze(0) + b.unsqueeze(1)` `loss = out.sigmoid().prod()` `loss.backward()`

Detailed Documentation

Documentation

A simple lookup table that stores embeddings of a fixed dictionary and size.

This module is often used to store word embeddings and retrieve them using indices.

The input to the module is a list of indices, and the output is the corresponding word embeddings.

`num_embeddings` (int) – size of the dictionary of embeddings

`embedding_dim` (int) – the size of each embedding vector

`padding_idx` (int, optional) – If specified, the entries at `padding_idx` do not contribute to the gradient; therefore, the embedding vector at `padding_idx` is not updated during training, i.e. it remains as a fixed “pad”. For a newly constructed Embedding, the embedding vector at `padding_idx` will default to all zeros, but can be updated to another value to be used as the padding vector.

`max_norm` (float, optional) – If given, each embedding vector with norm larger than `max_norm` is renormalized to have norm `max_norm`.

`norm_type` (float, optional) – The p of the p -norm to compute for the `max_norm` option. Default 2.

`scale_grad_by_freq` (bool, optional) – If given, this will scale gradients by the inverse of frequency of the words in the mini-batch. Default False.

`sparse` (bool, optional) – If True, gradient w.r.t. weight matrix will be a sparse tensor. See Notes for more details regarding sparse gradients.

`weight` (Tensor) – the learnable weights of the module of shape $(\text{num_embeddings}, \text{embedding_dim})$ initialized from $\mathcal{N}(0, 1)$

Input: $(*)^*(*)$, `IntTensor` or `LongTensor` of arbitrary shape containing the indices to extract

Output: $(*, H)^{(*)}$, $H(*, H)$, where $*$ is the input shape and

`H=embedding_dim`
`H=\text{embedding\_dim}`
`H=embedding_dim`

Keep in mind that only a limited number of optimizers support sparse gradients: currently it's `optim.SGD` (CUDA and CPU), `optim.SparseAdam` (CUDA and CPU) and `optim.Adagrad` (CPU)

When `max_norm` is not `None`, `Embedding`'s forward method will modify the weight tensor in-place. Since tensors needed for gradient computations cannot be modified in-place, performing a differentiable operation on `Embedding.weight` before calling `Embedding`'s forward method requires cloning `Embedding.weight` when `max_norm` is not `None`. For example:

Create `Embedding` instance from given 2-dimensional `FloatTensor`.

`embeddings` (`Tensor`) – `FloatTensor` containing weights for the `Embedding`. First dimension is being passed to `Embedding` as `num_embeddings`, second as `embedding_dim`.

`freeze` (`bool`, optional) – If `True`, the tensor does not get updated in the learning process. Equivalent to `embedding.weight.requires_grad = False`. Default: `True`

`padding_idx` (`int`, optional) – If specified, the entries at `padding_idx` do not contribute to the gradient; therefore, the embedding vector at `padding_idx` is not updated during training, i.e. it remains as a fixed “pad”.

`max_norm` (`float`, optional) – See module initialization documentation.

`norm_type` (`float`, optional) – See module initialization documentation. Default 2.

`scale_grad_by_freq` (`bool`, optional) – See module initialization documentation. Default `False`.

sparse (bool, optional) – See module initialization documentation.